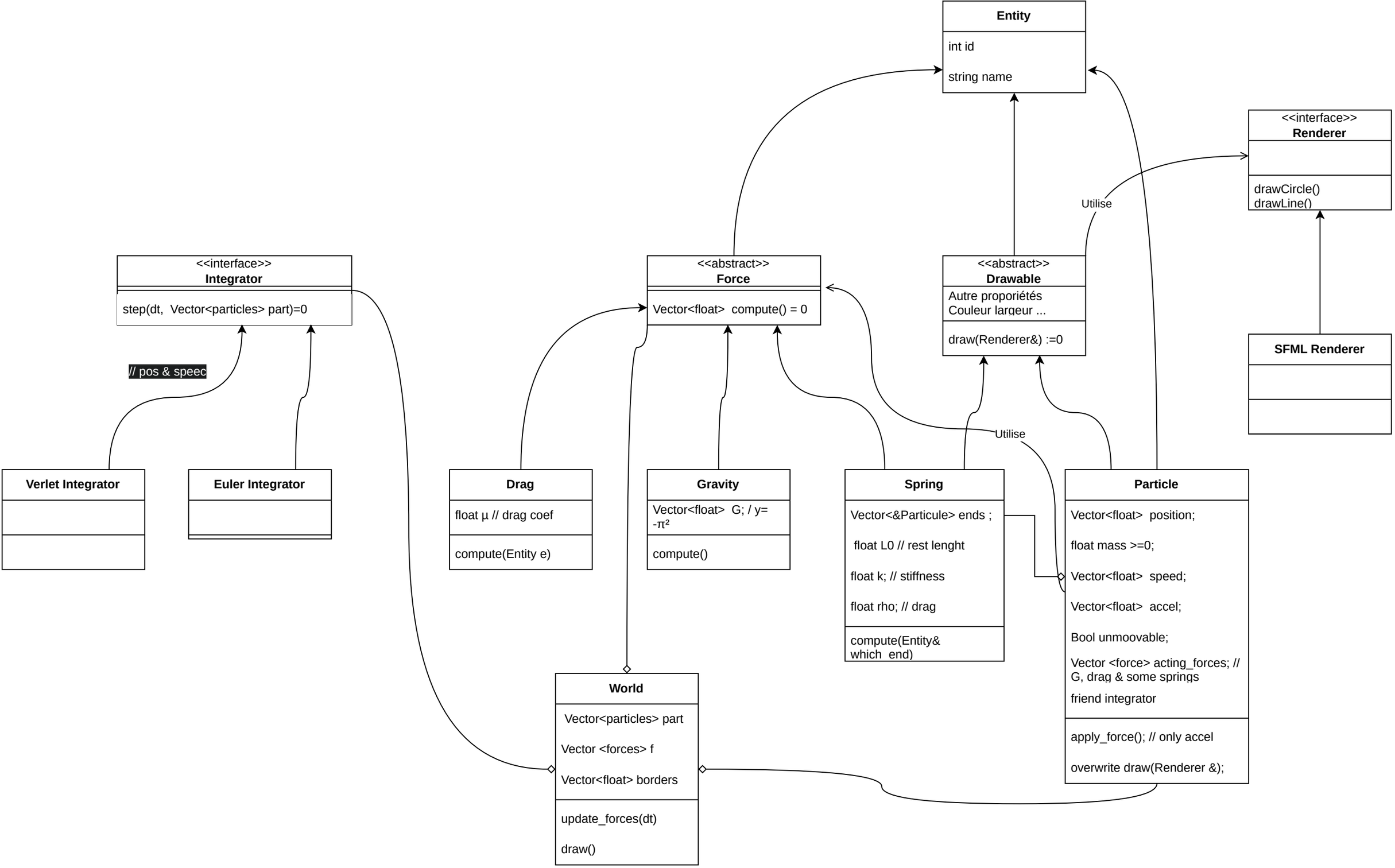
Entity
int id
string name
<<interface>>
Renderer
drawCircle()
drawLine()
Utilise
<<interface>>
Integrator
step(dt, Vector<particles> part)=0
<<abstract>>
Force
Vector<float> compute() = 0
<<abstract>>
Drawable
Autre proporiétés
Couleur largeur ...
draw(Renderer&) :=0
SFML Renderer
// pos & speec
Utilise
Verlet Integrator
Euler Integrator
Drag
float μ // drag coef
compute(Entity e)
Gravity
Vector<float> G; / y= -π²
compute()
Spring
Vector<&Particule> ends ;
float L0 // rest lenght
float k; // stiffness
float rho; // drag
compute(Entity& which_end)
Particle
Vector<float> position;
float mass >=0;
Vector<float> speed;
Vector<float> accel;
Bool unmoovable;
Vector <force> acting_forces; // G, drag & some springs
friend integrator
apply_force(); // only accel
overwrite draw(Renderer &);
World
Vector<particles> part
Vector <forces> f
Vector<float> borders
update_forces(dt)
draw()